UNICEUB · CIENCIA DA COMPUTACAO · INTEGRACAO DEVOPS

LABORATORIO DE CONTAINERS COM DOCKER

Unidade 03 · Registro de execucao com evidencias coletadas automaticamente

Situacao	LABORATORIO EXECUTADO
Data	28/10/2026 09:52:15
Projeto local	C:\Users\whiteracoon621\Documents\CEUB\DevOps-Unidade3\Reexecucoes\Reexecucao-20261028-094943\DevOpsWebApi
Repositorio	Repositorio GitHub nao publicado automaticamente. Consulte README e log.
Imagens previstas	devops-webapi:1.0.0, 1.1.0 e 1.2.0 (validacao individual nas evidencias)
Portas	8080 (inicial), 8081 (staging), 8082 (production), vinculadas a 127.0.0.1

Resultados e comprovantes

Dados reais produzidos pela execucao. HTTP 500 da release 1.2.0 e falha proposital e esperada. Consulte tambem os logs completos, arquivos Dockerfile, .dockerignore, compose.yaml e README.md.

Diretorio de execucao

OK

Preparar pastas de trabalho protegidas · 2026-10-28 09:49:48

```
C:\Users\whiteracoon621\Documents\CEUB\DevOps-Unidade3\Reexecucoes\Reexecucao-20261028-094943\DevOpsWebApi
```

Docker CLI

OK

Validar Docker Engine e Compose (sem UAC) · 2026-10-28 09:49:48

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe --version
Docker version 29.5.2, build 79eb04c
```

Docker Engine

OK

Validar Docker Engine e Compose (sem UAC) · 2026-10-28 09:49:49

```
Server version: 29.5.2
```

Docker Compose V2

OK

Validar Docker Engine e Compose (sem UAC) · 2026-10-28 09:49:50

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe compose version
Docker Compose version v5.1.3
```

SDK .NET utilizado

OK

Preparar .NET 10 sem administrador · 2026-10-28 09:49:51

```
C:\Program Files\dotnet\dotnet.exe --version
10.0.300
```

docker run --rm hello-world

OK

Testar o Docker com hello-world · 2026-10-28 09:49:55

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe run --rm hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

Criar template Web API .NET 10

OK

Gerar a API ASP.NET Core e os arquivos de build · 2026-10-28 09:49:56

```
C:\Program Files\dotnet\dotnet.exe new webapi -n DevOpsWebApi -o C:\Users\whiteracoon621\Documents\CEUB\DevOps-
Unidade3\Reexecucoes\Reexecucao-20261028-094943\DevOpsWebApi --framework net10.0 --no-restore
O modelo "API Web do ASP.NET Core" foi criado com êxito.
```

Restauracao de dependencias do projeto local

OK

Gerar a API ASP.NET Core e os arquivos de build · 2026-10-28 09:49:59

```
C:\Program Files\dotnet\dotnet.exe restore C:\Users\whiteracoon621\Documents\CEUB\DevOps-Unidade3\Reexecucoes\Reexecucao-
20261028-094943\DevOpsWebApi\DevOpsWebApi.csproj
  Determinando os projetos a serem restaurados...
  C:\Users\whiteracoon621\Documents\CEUB\DevOps-Unidade3\Reexecucoes\Reexecucao-20261028-
094943\DevOpsWebApi\DevOpsWebApi.csproj restaurado (em 150 ms).
```

Build local da API 1.0.0

OK

Gerar a API ASP.NET Core e os arquivos de build · 2026-10-28 09:50:05

```
C:\Program Files\dotnet\dotnet.exe build C:\Users\whiteracoon621\Documents\CEUB\DevOps-Unidade3\Reexecucoes\Reexecucao-
20261028-094943\DevOpsWebApi\DevOpsWebApi.csproj -c Release --no-restore
  DevOpsWebApi -> C:\Users\whiteracoon621\Documents\CEUB\DevOps-Unidade3\Reexecucoes\Reexecucao-20261028-
094943\DevOpsWebApi\bin\Release\net10.0\DevOpsWebApi.dll

Compilação com êxito.
    0 Aviso(s)
    0 Erro(s)

Tempo Decorrido 00:00:04.34
```

Verificacao HTTP http://127.0.0.1:5096/

OK

Validar os endpoints da API 1.0.0 sem Docker · 2026-10-28 09:50:06

```
GET http://127.0.0.1:5096/
HTTP 200 (esperado 200)
{"application":"DevOps Web API","version":"1.0.0","environment":"development"}
```

Verificacao HTTP http://127.0.0.1:5096/health

OK

Validar os endpoints da API 1.0.0 sem Docker · 2026-10-28 09:50:06

```
GET http://127.0.0.1:5096/health
HTTP 200 (esperado 200)
{"status":"healthy","version":"1.0.0","environment":"development","timestamp":"2026-10-28T12:50:06.509129Z"}
```

Multi-stage Docker build da imagem 1.0.0

OK

Construir imagem versionada 1.0.0 · 2026-10-28 09:50:23

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe build -t devops-webapi:1.0.0
C:\Users\whiteracoon621\Documents\CEUB\DevOps-Unidade3\Reexecucoes\Reexecucao-20261028-094943\DevOpsWebApi
#0 building with "desktop-linux" instance using docker driver

#1 [internal] load build definition from Dockerfile
#1 transferring dockerfile: 461B done
#1 DONE 0.0s

#2 [internal] load metadata for mcr.microsoft.com/dotnet/aspnet:10.0
#2 ...

#3 [internal] load metadata for mcr.microsoft.com/dotnet/sdk:10.0
#3 DONE 0.5s

#2 [internal] load metadata for mcr.microsoft.com/dotnet/aspnet:10.0
#2 DONE 0.5s

#4 [internal] load .dockerignore
#4 transferring context: 113B done
#4 DONE 0.0s

#5 [internal] load build context
#5 DONE 0.0s

#6 [runtime 1/3] FROM
mcr.microsoft.com/dotnet/aspnet:10.0@sha256:de0ee358cdd672f7017a003bed21e9ac9ed48a27c922a59b18175b6e82fa7fee
#6 resolve mcr.microsoft.com/dotnet/aspnet:10.0@sha256:de0ee358cdd672f7017a003bed21e9ac9ed48a27c922a59b18175b6e82fa7fee
0.1s done
#6 DONE 0.1s

#7 [build 1/6] FROM
mcr.microsoft.com/dotnet/sdk:10.0@sha256:de4fe90a98845d75910b9769662465c84de1a934665c431279d1382b2bddc02b
#7 resolve mcr.microsoft.com/dotnet/sdk:10.0@sha256:de4fe90a98845d75910b9769662465c84de1a934665c431279d1382b2bddc02b 0.2s
done
#7 DONE 0.2s

#5 [internal] load build context
#5 transferring context: 3.46kB 0.1s done
#5 DONE 0.2s

#8 [build 2/6] WORKDIR /src
#8 CACHED

#9 [build 3/6] COPY DevOpsWebApi.csproj .
#9 CACHED

#10 [build 4/6] RUN dotnet restore
#10 CACHED

#11 [build 5/6] COPY . .
#11 DONE 0.2s

#12 [build 6/6] RUN dotnet publish -c Release -o /app/publish --no-restore
#12 7.026   DevOpsWebApi -> /src/bin/Release/net10.0/DevOpsWebApi.dll
#12 7.083   DevOpsWebApi -> /app/publish/
#12 DONE 7.2s

#13 [runtime 2/3] WORKDIR /app
#13 CACHED

#14 [runtime 3/3] COPY --from=build /app/publish .
#14 CACHED
```

```
#15 exporting to image
#15 exporting layers done
#15 exporting manifest sha256:00c73b5ce555baa8d319820461968d67d8b455a7825e75ada63e07990a705574 0.0s done
#15 exporting config sha256:1757981d9e077056575430f94cea75365c1d4c1cd6987488cbce656ed88a0465 0.0s done
#15 exporting attestation manifest sha256:858afa679e60818ffe90cde99b9bb0bcc28908b2b23f6b55f385c563022950c0
#15 exporting attestation manifest sha256:858afa679e60818ffe90cde99b9bb0bcc28908b2b23f6b55f385c563022950c0 0.1s done
#15 exporting manifest list sha256:32e9d354120331ac442a5e6ae20851ce628bdcd8111e4e49f94e3ac49b2e2cd4 0.0s done
#15 naming to docker.io/library/devops-webapi:1.0.0 done
#15 unpacking to docker.io/library/devops-webapi:1.0.0 0.1s done
#15 DONE 0.3s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/ykfe87spbbjre93et4x65wxm4
```

Imagem Docker 1.0.0 criada

OK

Construir imagem versionada 1.0.0 · 2026-10-28 09:50:23

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe image ls devops-webapi --format {{.Repository}}:{{.Tag}} {{.ID}}
{{.Size}}
devops-webapi:1.0.0 32e9d3541203 342MB
```

Deploy devops-api (development, imagem 1.0.0)

OK

Rodar, testar, inspecionar, parar e remover primeiro container · 2026-10-28 09:50:24

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe run -d --name devops-api --label ceub.devops.lab=ceub-devops-
unidade3 -p 127.0.0.1:8080:8080 -e APP_ENVIRONMENT=development -e APP_VERSION=1.0.0 devops-webapi:1.0.0
6f14e7fab73f935330487c547268114a9fe627898ae09a8d5a365b388cf0be56
```

Verificacao HTTP http://127.0.0.1:8080/

OK

Rodar, testar, inspecionar, parar e remover primeiro container · 2026-10-28 09:50:25

```
GET http://127.0.0.1:8080/
HTTP 200 (esperado 200)
{"application":"DevOps Web API","version":"1.0.0","environment":"development"}
```

Verificacao HTTP http://127.0.0.1:8080/health

OK

Rodar, testar, inspecionar, parar e remover primeiro container · 2026-10-28 09:50:25

```
GET http://127.0.0.1:8080/health
HTTP 200 (esperado 200)
{"status":"healthy","version":"1.0.0","environment":"development","timestamp":"2026-10-28T12:50:25.5400363Z"}
```

curl.exe -f http://127.0.0.1:8080/health

OK

Rodar, testar, inspecionar, parar e remover primeiro container · 2026-10-28 09:50:26

```
exit code: 0
{"status":"healthy","version":"1.0.0","environment":"development","timestamp":"2026-10-28T12:50:25.6719955Z"}
```

Container inicial em execucao (docker ps)

OK

Rodar, testar, inspecionar, parar e remover primeiro container · 2026-10-28 09:50:29

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe ps --filter name=devops-api
CONTAINER ID   IMAGE                 COMMAND                  CREATED         STATUS         PORTS
NAMES
6f14e7fab73f   devops-webapi:1.0.0   "dotnet DevOpsWebApi…"   6 seconds ago   Up 5 seconds   127.0.0.1:8080->8080/tcp
devops-api
```

Logs do container inicial

OK

Rodar, testar, inspecionar, parar e remover primeiro container · 2026-10-28 09:50:31

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe logs devops-api
warn: Microsoft.AspNetCore.Hosting.Diagnostics[15]
      Overriding HTTP_PORTS '8080' and HTTPS_PORTS ''. Binding to values defined by URLS instead 'http://+:8080'.
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://[::]:8080
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Production
info: Microsoft.Hosting.Lifetime[0]
      Content root path: /app
```

Inspecao do primeiro container

OK

Rodar, testar, inspecionar, parar e remover primeiro container · 2026-10-28 09:50:32

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe inspect -f {{.Name}} | {{.Config.Image}} | {{.State.Status}} |
{{.NetworkSettings.Ports}} devops-api
/devops-api | devops-webapi:1.0.0 | running | map[8080/tcp:[{127.0.0.1 8080}]]
```

Parar sem remover imagem

OK

Rodar, testar, inspecionar, parar e remover primeiro container · 2026-10-28 09:50:33

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe stop devops-api
devops-api
```

Container parado ainda existe

OK

Rodar, testar, inspecionar, parar e remover primeiro container · 2026-10-28 09:50:33

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe ps -a --filter name=devops-api
CONTAINER ID   IMAGE                 COMMAND                  CREATED          STATUS                     PORTS     NAMES
6f14e7fab73f   devops-webapi:1.0.0   "dotnet DevOpsWebApi…"   10 seconds ago   Exited (0) 1 second ago              devops-
api
```

Remocao segura do container do laboratorio devops-api

OK

Rodar, testar, inspecionar, parar e remover primeiro container · 2026-10-28 09:50:34

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe rm -f devops-api
devops-api
```

Deploy devops-api-staging (staging, imagem 1.0.0)

OK

Implantar 1.0.0 em staging e production · 2026-10-28 09:50:35

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe run -d --name devops-api-staging --label ceub.devops.lab=ceub-
devops-unidade3 -p 127.0.0.1:8081:8080 -e APP_ENVIRONMENT=staging -e APP_VERSION=1.0.0 devops-webapi:1.0.0
81d1aa9bbf3ac2fd51894a5265e8db5894c0704df886f2599696a2682862d21b
```

Deploy devops-api-prod (production, imagem 1.0.0)

OK

Implantar 1.0.0 em staging e production · 2026-10-28 09:50:36

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe run -d --name devops-api-prod --label ceub.devops.lab=ceub-devops-
unidade3 -p 127.0.0.1:8082:8080 -e APP_ENVIRONMENT=production -e APP_VERSION=1.0.0 devops-webapi:1.0.0
1b118a965bab1f9b10e2a3312dc492a5a81060ef18920494e2b226b1ff644143
```

Verificacao HTTP http://127.0.0.1:8081/

OK

Implantar 1.0.0 em staging e production · 2026-10-28 09:50:36

```
GET http://127.0.0.1:8081/
HTTP 200 (esperado 200)
{"application":"DevOps Web API","version":"1.0.0","environment":"staging"}
```

Verificacao HTTP http://127.0.0.1:8082/

OK

Implantar 1.0.0 em staging e production · 2026-10-28 09:50:37

```
GET http://127.0.0.1:8082/
HTTP 200 (esperado 200)
{"application":"DevOps Web API","version":"1.0.0","environment":"production"}
```

Staging e production 1.0.0 simultaneos

OK

Implantar 1.0.0 em staging e production · 2026-10-28 09:50:41

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe ps --filter name=devops-api
CONTAINER ID   IMAGE                 COMMAND                  CREATED         STATUS         PORTS
NAMES
1b118a965bab   devops-webapi:1.0.0   "dotnet DevOpsWebApi…"   5 seconds ago   Up 4 seconds   127.0.0.1:8082->8080/tcp
devops-api-prod
81d1aa9bbf3a   devops-webapi:1.0.0   "dotnet DevOpsWebApi…"   6 seconds ago   Up 5 seconds   127.0.0.1:8081->8080/tcp
devops-api-staging
```

Build local da release 1.1.0

OK

Adicionar endpoint /version e testar 1.1.0 localmente · 2026-10-28 09:50:44

```
C:\Program Files\dotnet\dotnet.exe build C:\Users\whiteracoon621\Documents\CEUB\DevOps-Unidade3\Reexecucoes\Reexecucao-
20261028-094943\DevOpsWebApi\DevOpsWebApi.csproj -c Release --no-restore
  DevOpsWebApi -> C:\Users\whiteracoon621\Documents\CEUB\DevOps-Unidade3\Reexecucoes\Reexecucao-20261028-
094943\DevOpsWebApi\bin\Release\net10.0\DevOpsWebApi.dll

Compilação com êxito.
    0 Aviso(s)
    0 Erro(s)

Tempo Decorrido 00:00:01.29
```

Verificacao HTTP http://127.0.0.1:5096/version

OK

Adicionar endpoint /version e testar 1.1.0 localmente · 2026-10-28 09:50:46

```
GET http://127.0.0.1:5096/version
HTTP 200 (esperado 200)
{"version":"1.1.0","environment":"development","message":"Nova funcionalidade da versao 1.1"}
```

Imagem Docker 1.1.0

OK

Construir imagem 1.1.0 sem descartar 1.0.0 · 2026-10-28 09:50:57

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe build -t devops-webapi:1.1.0
C:\Users\whiteracoon621\Documents\CEUB\DevOps-Unidade3\Reexecucoes\Reexecucao-20261028-094943\DevOpsWebApi
#0 building with "desktop-linux" instance using docker driver

#1 [internal] load build definition from Dockerfile
#1 transferring dockerfile: 461B done
#1 DONE 0.0s

#2 [internal] load metadata for mcr.microsoft.com/dotnet/sdk:10.0
#2 ...

#3 [internal] load metadata for mcr.microsoft.com/dotnet/aspnet:10.0
#3 DONE 0.3s

#2 [internal] load metadata for mcr.microsoft.com/dotnet/sdk:10.0
#2 DONE 0.3s

#4 [internal] load .dockerignore
#4 transferring context: 113B done
#4 DONE 0.0s

#5 [runtime 1/3] FROM
mcr.microsoft.com/dotnet/aspnet:10.0@sha256:de0ee358cdd672f7017a003bed21e9ac9ed48a27c922a59b18175b6e82fa7fee
#5 resolve mcr.microsoft.com/dotnet/aspnet:10.0@sha256:de0ee358cdd672f7017a003bed21e9ac9ed48a27c922a59b18175b6e82fa7fee
0.0s done
#5 DONE 0.0s

#6 [build 1/6] FROM
mcr.microsoft.com/dotnet/sdk:10.0@sha256:de4fe90a98845d75910b9769662465c84de1a934665c431279d1382b2bddc02b
#6 resolve mcr.microsoft.com/dotnet/sdk:10.0@sha256:de4fe90a98845d75910b9769662465c84de1a934665c431279d1382b2bddc02b 0.0s
done
#6 DONE 0.1s

#7 [internal] load build context
#7 transferring context: 1.55kB done
#7 DONE 0.0s

#8 [build 3/6] COPY DevOpsWebApi.csproj .
#8 CACHED

#9 [build 2/6] WORKDIR /src
#9 CACHED

#10 [build 4/6] RUN dotnet restore
#10 CACHED

#11 [build 5/6] COPY . .
#11 DONE 0.0s

#12 [build 6/6] RUN dotnet publish -c Release -o /app/publish --no-restore
#12 4.795   DevOpsWebApi -> /src/bin/Release/net10.0/DevOpsWebApi.dll
#12 4.950   DevOpsWebApi -> /app/publish/
#12 DONE 5.1s

#13 [runtime 2/3] WORKDIR /app
#13 CACHED

#14 [runtime 3/3] COPY --from=build /app/publish .
#14 CACHED

#15 exporting to image
#15 exporting layers done
```

```
#15 exporting manifest sha256:46a60540b55e1009c4aec232c3098620eac021360132afa55945896d93dc170f done
#15 exporting config sha256:6f45a8cce0299c0a72704df64f3d174c591902eee0f63b17fc45f0e301143582 done
#15 exporting attestation manifest sha256:86a7a2baa2b63c1248bdf6b3af83f701988ec57d0602a402e43c067fab8f6286 0.1s done
#15 exporting manifest list sha256:cf4728fe3520f226c9199d4522be487520cdc7087818a6867506f462d1acb6f3
#15 exporting manifest list sha256:cf4728fe3520f226c9199d4522be487520cdc7087818a6867506f462d1acb6f3 0.1s done
#15 naming to docker.io/library/devops-webapi:1.1.0
#15 naming to docker.io/library/devops-webapi:1.1.0 0.0s done
#15 unpacking to docker.io/library/devops-webapi:1.1.0 0.1s done
#15 DONE 0.4s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/urj4yw594vxi1jzzwwkb30tc0
```

Imagens 1.0.0 e 1.1.0 simultaneas

OK

Construir imagem 1.1.0 sem descartar 1.0.0 · 2026-10-28 09:50:58

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe image ls devops-webapi --format {{.Repository}}:{{.Tag}} {{.ID}}
{{.Size}}
devops-webapi:1.1.0 cf4728fe3520 342MB
devops-webapi:1.0.0 32e9d3541203 342MB
```

Remocao segura do container do laboratorio devops-api-staging

OK

Promover a versao 1.1.0 para staging e validar gate · 2026-10-28 09:50:59

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe rm -f devops-api-staging
devops-api-staging
```

Deploy devops-api-staging (staging, imagem 1.1.0)

OK

Promover a versao 1.1.0 para staging e validar gate · 2026-10-28 09:51:00

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe run -d --name devops-api-staging --label ceub.devops.lab=ceub-
devops-unidade3 -p 127.0.0.1:8081:8080 -e APP_ENVIRONMENT=staging -e APP_VERSION=1.1.0 devops-webapi:1.1.0
0500806e4eca1a2c6a7f1c7414b73dfd493fbb1717ad29fcb501059600704e24
```

Verificacao HTTP http://127.0.0.1:8081/health

OK

Promover a versao 1.1.0 para staging e validar gate · 2026-10-28 09:51:00

```
GET http://127.0.0.1:8081/health
HTTP 200 (esperado 200)
{"status":"healthy","version":"1.1.0","environment":"staging","timestamp":"2026-10-28T12:51:00.4084463Z"}
```

Verificacao HTTP http://127.0.0.1:8081/version

OK

Promover a versao 1.1.0 para staging e validar gate · 2026-10-28 09:51:00

```
GET http://127.0.0.1:8081/version
HTTP 200 (esperado 200)
{"version":"1.1.0","environment":"staging","message":"Nova funcionalidade da versao 1.1"}
```

curl.exe -f http://127.0.0.1:8081/health

OK

Promover a versao 1.1.0 para staging e validar gate · 2026-10-28 09:51:01

```
exit code: 0
{"status":"healthy","version":"1.1.0","environment":"staging","timestamp":"2026-10-28T12:51:00.6611255Z"}
```

Verificacao HTTP http://127.0.0.1:8082/health

OK

Promover a versao 1.1.0 para staging e validar gate · 2026-10-28 09:51:01

```
GET http://127.0.0.1:8082/health
HTTP 200 (esperado 200)
{"status":"healthy","version":"1.1.0","environment":"production","timestamp":"2026-10-28T12:51:01.0471641Z"}
```

Quality Gate em staging

OK

Promover a versao 1.1.0 para staging e validar gate · 2026-10-28 09:51:01

```
Release 1.1.0 aprovada; production continua 1.0.0 ate a promocao.
```

Remocao segura do container do laboratorio devops-api-prod

OK

Promover imagem validada 1.1.0 para production · 2026-10-28 09:51:06

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe rm -f devops-api-prod
devops-api-prod
```

Deploy devops-api-prod (production, imagem 1.1.0)

OK

Promover imagem validada 1.1.0 para production · 2026-10-28 09:51:07

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe run -d --name devops-api-prod --label ceub.devops.lab=ceub-devops-
unidade3 -p 127.0.0.1:8082:8080 -e APP_ENVIRONMENT=production -e APP_VERSION=1.1.0 devops-webapi:1.1.0
bf4952a4e444d599a0f7f64e62f4f101a257f35777df013a1025149c45a98bd5
```

Verificacao HTTP http://127.0.0.1:8082/health

OK

Promover imagem validada 1.1.0 para production · 2026-10-28 09:51:07

```
GET http://127.0.0.1:8082/health
HTTP 200 (esperado 200)
{"status":"healthy","version":"1.1.0","environment":"production","timestamp":"2026-10-28T12:51:07.1660912Z"}
```

Verificacao HTTP http://127.0.0.1:8082/version

OK

Promover imagem validada 1.1.0 para production · 2026-10-28 09:51:07

```
GET http://127.0.0.1:8082/version
HTTP 200 (esperado 200)
{"version":"1.1.0","environment":"production","message":"Nova funcionalidade da versao 1.1"}
```

curl.exe -f http://127.0.0.1:8082/health

OK

Promover imagem validada 1.1.0 para production · 2026-10-28 09:51:07

```
exit code: 0
{"status":"healthy","version":"1.1.0","environment":"production","timestamp":"2026-10-28T12:51:07.3995984Z"}
```

Staging e production na release 1.1.0

OK

Promover imagem validada 1.1.0 para production · 2026-10-28 09:51:08

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe ps --filter name=devops-api
CONTAINER ID   IMAGE                 COMMAND                  CREATED         STATUS         PORTS
NAMES
bf4952a4e444   devops-webapi:1.1.0   "dotnet DevOpsWebApi…"   1 second ago    Up 1 second    127.0.0.1:8082->8080/tcp
devops-api-prod
0500806e4eca   devops-webapi:1.1.0   "dotnet DevOpsWebApi…"   8 seconds ago   Up 8 seconds   127.0.0.1:8081->8080/tcp
devops-api-staging
```

Imagem 1.2.0 defeituosa para teste didatico

OK

Criar imagem 1.2.0 com HTTP 500 proposital · 2026-10-28 09:51:23

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe build -t devops-webapi:1.2.0
C:\Users\whiteracoon621\Documents\CEUB\DevOps-Unidade3\Reexecucoes\Reexecucao-20261028-094943\DevOpsWebApi
#0 building with "desktop-linux" instance using docker driver

#1 [internal] load build definition from Dockerfile
#1 transferring dockerfile: 461B 0.0s done
#1 DONE 0.0s

#2 [internal] load metadata for mcr.microsoft.com/dotnet/aspnet:10.0
#2 DONE 0.2s

#3 [internal] load metadata for mcr.microsoft.com/dotnet/sdk:10.0
#3 DONE 0.2s

#4 [internal] load .dockerignore
#4 transferring context: 113B 0.0s done
#4 DONE 0.0s

#5 [internal] load build context
#5 DONE 0.0s

#6 [build 1/6] FROM
mcr.microsoft.com/dotnet/sdk:10.0@sha256:de4fe90a98845d75910b9769662465c84de1a934665c431279d1382b2bddc02b
#6 resolve mcr.microsoft.com/dotnet/sdk:10.0@sha256:de4fe90a98845d75910b9769662465c84de1a934665c431279d1382b2bddc02b
#6 resolve mcr.microsoft.com/dotnet/sdk:10.0@sha256:de4fe90a98845d75910b9769662465c84de1a934665c431279d1382b2bddc02b 0.1s
done
#6 DONE 0.2s

#7 [runtime 1/3] FROM
mcr.microsoft.com/dotnet/aspnet:10.0@sha256:de0ee358cdd672f7017a003bed21e9ac9ed48a27c922a59b18175b6e82fa7fee
#7 resolve mcr.microsoft.com/dotnet/aspnet:10.0@sha256:de0ee358cdd672f7017a003bed21e9ac9ed48a27c922a59b18175b6e82fa7fee
0.2s done
#7 DONE 0.2s

#5 [internal] load build context
#5 transferring context: 1.51kB 0.0s done
#5 DONE 0.0s

#8 [build 2/6] WORKDIR /src
#8 CACHED

#9 [build 3/6] COPY DevOpsWebApi.csproj .
#9 CACHED

#10 [build 4/6] RUN dotnet restore
#10 CACHED

#11 [build 5/6] COPY . .
#11 DONE 0.1s

#12 [build 6/6] RUN dotnet publish -c Release -o /app/publish --no-restore
#12 4.854   DevOpsWebApi -> /src/bin/Release/net10.0/DevOpsWebApi.dll
#12 4.912   DevOpsWebApi -> /app/publish/
#12 DONE 5.0s

#13 [runtime 2/3] WORKDIR /app
#13 CACHED

#14 [runtime 3/3] COPY --from=build /app/publish .
#14 CACHED

#15 exporting to image
```

```
#15 exporting layers done
#15 exporting manifest sha256:f72667da69ebae68e4d2fd812dca2b26582a8df0672816caaa8362ad7cdb9190 0.0s done
#15 exporting config sha256:1d3a9877c2ebbd0581c48cfaca42ca9fba52ba5233547b031051b3d88e099c1b done
#15 exporting attestation manifest sha256:b9375b965d22932059ea29ea840f53ff5b9170696551a76024d838e291188ba7
#15 exporting attestation manifest sha256:b9375b965d22932059ea29ea840f53ff5b9170696551a76024d838e291188ba7 0.1s done
#15 exporting manifest list sha256:c259cbb5f56626db3c38d82d03f012ca43afd6f54b6b4202fd15ee436b4b7dc6 0.0s done
#15 naming to docker.io/library/devops-webapi:1.2.0 done
#15 unpacking to docker.io/library/devops-webapi:1.2.0 0.0s done
#15 DONE 0.2s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/qanklxph00w1rlwbcu7j1gqei
```

Evidencia das tres imagens versionadas

OK

Criar imagem 1.2.0 com HTTP 500 proposital · 2026-10-28 09:51:24

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe image ls devops-webapi --format {{.Repository}}:{{.Tag}} {{.ID}}
{{.Size}}
devops-webapi:1.2.0 c259cbb5f566 342MB
devops-webapi:1.1.0 cf4728fe3520 342MB
devops-webapi:1.0.0 32e9d3541203 342MB
```

Remocao segura do container do laboratorio devops-api-staging

OK

Testar 1.2.0 em staging e BLOQUEAR promocao normal · 2026-10-28 09:51:25

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe rm -f devops-api-staging
devops-api-staging
```

Deploy devops-api-staging (staging, imagem 1.2.0)

OK

Testar 1.2.0 em staging e BLOQUEAR promocao normal · 2026-10-28 09:51:26

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe run -d --name devops-api-staging --label ceub.devops.lab=ceub-
devops-unidade3 -p 127.0.0.1:8081:8080 -e APP_ENVIRONMENT=staging -e APP_VERSION=1.2.0 devops-webapi:1.2.0
e38105f2f97d30135295745aad8f31f72e0c7ef316a8b959cff4a60ded3ed4e7
```

Verificacao HTTP http://127.0.0.1:8081/health

FALHA ESPERADA (DIDATICA)

Testar 1.2.0 em staging e BLOQUEAR promocao normal · 2026-10-28 09:51:26

```
GET http://127.0.0.1:8081/health
HTTP 500 (esperado 500)

Diagnostico: O servidor remoto retornou um erro: (500) Erro Interno do Servidor.
```

curl.exe -f http://127.0.0.1:8081/health

FALHA ESPERADA (DIDATICA)

Testar 1.2.0 em staging e BLOQUEAR promocao normal · 2026-10-28 09:51:27

```
exit code: 22
curl: (22) The requested URL returned error: 500
```

Quality Gate da release 1.2.0

FALHA ESPERADA (DIDATICA)

Testar 1.2.0 em staging e BLOQUEAR promocao normal · 2026-10-28 09:51:27

```
HTTP 500 confirmado: uma pipeline real nao promoveria esta imagem.
```

Verificacao HTTP http://127.0.0.1:8082/health

OK

Testar 1.2.0 em staging e BLOQUEAR promocao normal · 2026-10-28 09:51:27

```
GET http://127.0.0.1:8082/health
HTTP 200 (esperado 200)
{"status":"healthy","version":"1.1.0","environment":"production","timestamp":"2026-10-28T12:51:27.0840872Z"}
```

Remocao segura do container do laboratorio devops-api-prod

OK

Simular incidente APENAS em production local de laboratorio · 2026-10-28 09:51:32

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe rm -f devops-api-prod
devops-api-prod
```

Deploy devops-api-prod (production, imagem 1.2.0)

OK

Simular incidente APENAS em production local de laboratorio · 2026-10-28 09:51:33

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe run -d --name devops-api-prod --label ceub.devops.lab=ceub-devops-unidade3 -p 127.0.0.1:8082:8080 -e APP_ENVIRONMENT=production -e APP_VERSION=1.2.0 devops-webapi:1.2.0
d74454d01dc3421411568da3bc80b1b5b43a0ed1c5689bebe1bc0b0d2acad56c
```

Verificacao HTTP http://127.0.0.1:8082/health

FALHA ESPERADA (DIDATICA)

Simular incidente APENAS em production local de laboratorio · 2026-10-28 09:51:34

```
GET http://127.0.0.1:8082/health
HTTP 500 (esperado 500)

Diagnostico: O servidor remoto retornou um erro: (500) Erro Interno do Servidor.
```

curl.exe -f http://127.0.0.1:8082/health

FALHA ESPERADA (DIDATICA)

Simular incidente APENAS em production local de laboratorio · 2026-10-28 09:51:35

```
exit code: 22
curl: (22) The requested URL returned error: 500
```

Incidente simulado na production de laboratorio

FALHA ESPERADA (DIDATICA)

Simular incidente APENAS em production local de laboratorio · 2026-10-28 09:51:35

```
Production local com imagem 1.2.0 retornou HTTP 500.
```

Remocao segura do container do laboratorio devops-api-prod

OK

Executar rollback de production para 1.1.0 · 2026-10-28 09:51:40

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe rm -f devops-api-prod
devops-api-prod
```

Deploy devops-api-prod (production, imagem 1.1.0)

OK

Executar rollback de production para 1.1.0 · 2026-10-28 09:51:41

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe run -d --name devops-api-prod --label ceub.devops.lab=ceub-devops-unidade3 -p 127.0.0.1:8082:8080 -e APP_ENVIRONMENT=production -e APP_VERSION=1.1.0 devops-webapi:1.1.0
279ca5dc1e14f524905e72a7c7189a72bc14947fb3376bd93b4793684b56a300
```

Verificacao HTTP http://127.0.0.1:8082/health

OK

Executar rollback de production para 1.1.0 · 2026-10-28 09:51:42

```
GET http://127.0.0.1:8082/health
HTTP 200 (esperado 200)
{"status":"healthy","version":"1.1.0","environment":"production","timestamp":"2026-10-28T12:51:42.3233103Z"}
```

curl.exe -f http://127.0.0.1:8082/health

OK

Executar rollback de production para 1.1.0 · 2026-10-28 09:51:42

```
exit code: 0
{"status":"healthy","version":"1.1.0","environment":"production","timestamp":"2026-10-28T12:51:42.4453044Z"}
```

ROLLBACK CONCLUIDO

OK

Executar rollback de production para 1.1.0 · 2026-10-28 09:51:42

```
Imagem devops-webapi:1.1.0 restaurada em production; /health = HTTP 200.
```

Estado apos rollback (staging falho e production saudavel)

OK

Executar rollback de production para 1.1.0 · 2026-10-28 09:51:47

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe ps --filter name=devops-api
CONTAINER ID   IMAGE                 COMMAND                  CREATED          STATUS          PORTS
NAMES
279ca5dc1e14   devops-webapi:1.1.0   "dotnet DevOpsWebApi…"   6 seconds ago    Up 6 seconds    127.0.0.1:8082->8080/tcp
devops-api-prod
e38105f2f97d   devops-webapi:1.2.0   "dotnet DevOpsWebApi…"   21 seconds ago   Up 20 seconds   127.0.0.1:8081->8080/tcp
devops-api-staging
```

Validacao YAML do Docker Compose

OK

Criar compose.yaml com healthcheck sem curl · 2026-10-28 09:51:56

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe compose -f C:\Users\whiteracoon621\Documents\CEUB\DevOps-
Unidade3\Reexecucoes\Reexecucao-20261028-094943\DevOpsWebApi\compose.yaml config
name: ceub-devops-unidade3
services:
  production:
    container_name: devops-api-prod
    environment:
      APP_ENVIRONMENT: production
      APP_VERSION: 1.1.0
    healthcheck:
      test:
        - CMD
        - dotnet
        - DevOpsWebApi.dll
        - --health-check
      timeout: 5s
      interval: 5s
      retries: 4
      start_period: 5s
    image: devops-webapi:1.1.0
    labels:
      ceub.devops.lab: ceub-devops-unidade3
    networks:
      default: null
    ports:
      - mode: ingress
        host_ip: 127.0.0.1
        target: 8080
        published: "8082"
        protocol: tcp
  staging:
    container_name: devops-api-staging
    environment:
      APP_ENVIRONMENT: staging
      APP_VERSION: 1.1.0
    healthcheck:
      test:
        - CMD
        - dotnet
        - DevOpsWebApi.dll
        - --health-check
      timeout: 5s
      interval: 5s
      retries: 4
      start_period: 5s
    image: devops-webapi:1.1.0
    labels:
      ceub.devops.lab: ceub-devops-unidade3
    networks:
      default: null
    ports:
      - mode: ingress
        host_ip: 127.0.0.1
        target: 8080
        published: "8081"
        protocol: tcp
networks:
  default:
    name: ceub-devops-unidade3_default
```

Remocao segura do container do laboratorio devops-api-staging

OK

Criar compose.yaml com healthcheck sem curl · 2026-10-28 09:51:57

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe rm -f devops-api-staging
devops-api-staging
```

Remocao segura do container do laboratorio devops-api-prod

OK

Criar compose.yaml com healthcheck sem curl · 2026-10-28 09:51:58

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe rm -f devops-api-prod
devops-api-prod
```

docker compose up -d

OK

Criar compose.yaml com healthcheck sem curl · 2026-10-28 09:52:00

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe compose -f C:\Users\whiteracoon621\Documents\CEUB\DevOps-
Unidade3\Reexecucoes\Reexecucao-20261028-094943\DevOpsWebApi\compose.yaml up -d
 Network ceub-devops-unidade3_default Creating
 Network ceub-devops-unidade3_default Created
 Container devops-api-prod Creating
 Container devops-api-staging Creating
 Container devops-api-staging Created
 Container devops-api-prod Created
 Container devops-api-prod Starting
 Container devops-api-staging Starting
 Container devops-api-prod Started
 Container devops-api-staging Started
```

docker compose ps

OK

Criar compose.yaml com healthcheck sem curl · 2026-10-28 09:52:01

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe compose -f C:\Users\whiteracoon621\Documents\CEUB\DevOps-
Unidade3\Reexecucoes\Reexecucao-20261028-094943\DevOpsWebApi\compose.yaml ps
NAME                IMAGE                COMMAND                  SERVICE      CREATED        STATUS
PORTS
devops-api-prod     devops-webapi:1.1.0   "dotnet DevOpsWebApi…"   production   1 second ago   Up 1 second (health:
starting)   127.0.0.1:8082->8080/tcp
devops-api-staging   devops-webapi:1.1.0   "dotnet DevOpsWebApi…"   staging      1 second ago   Up 1 second (health:
starting)   127.0.0.1:8081->8080/tcp
```

Verificacao HTTP http://127.0.0.1:8081/health

OK

Criar compose.yaml com healthcheck sem curl · 2026-10-28 09:52:01

```
GET http://127.0.0.1:8081/health
HTTP 200 (esperado 200)
{"status":"healthy","version":"1.1.0","environment":"staging","timestamp":"2026-10-28T12:52:01.648063Z"}
```

Verificacao HTTP http://127.0.0.1:8082/health

OK

Criar compose.yaml com healthcheck sem curl · 2026-10-28 09:52:01

```
GET http://127.0.0.1:8082/health
HTTP 200 (esperado 200)
{"status":"healthy","version":"1.1.0","environment":"production","timestamp":"2026-10-28T12:52:01.8202841Z"}
```

Logs de staging e production pelo Compose

OK

Criar compose.yaml com healthcheck sem curl · 2026-10-28 09:52:05

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe compose -f C:\Users\whiteracoon621\Documents\CEUB\DevOps-
Unidade3\Reexecucoes\Reexecucao-20261028-094943\DevOpsWebApi\compose.yaml logs --tail 35
devops-api-staging  | warn: Microsoft.AspNetCore.Hosting.Diagnostics[15]
devops-api-staging  |       Overriding HTTP_PORTS '8080' and HTTPS_PORTS ''. Binding to values defined by URLS instead
'http://+:8080'.
devops-api-staging  | info: Microsoft.Hosting.Lifetime[14]
devops-api-staging  |       Now listening on: http://[::]:8080
devops-api-staging  | info: Microsoft.Hosting.Lifetime[0]
devops-api-staging  |       Application started. Press Ctrl+C to shut down.
devops-api-staging  | info: Microsoft.Hosting.Lifetime[0]
devops-api-staging  |       Hosting environment: Production
devops-api-staging  | info: Microsoft.Hosting.Lifetime[0]
devops-api-staging  |       Content root path: /app
devops-api-prod     | warn: Microsoft.AspNetCore.Hosting.Diagnostics[15]
devops-api-prod     |       Overriding HTTP_PORTS '8080' and HTTPS_PORTS ''. Binding to values defined by URLS instead
'http://+:8080'.
devops-api-prod     | info: Microsoft.Hosting.Lifetime[14]
devops-api-prod     |       Now listening on: http://[::]:8080
devops-api-prod     | info: Microsoft.Hosting.Lifetime[0]
devops-api-prod     |       Application started. Press Ctrl+C to shut down.
devops-api-prod     | info: Microsoft.Hosting.Lifetime[0]
devops-api-prod     |       Hosting environment: Production
devops-api-prod     | info: Microsoft.Hosting.Lifetime[0]
devops-api-prod     |       Content root path: /app
```

Docker HEALTHCHECK devops-api-staging

OK

Criar compose.yaml com healthcheck sem curl · 2026-10-28 09:52:05

```
State.Health.Status = healthy; checagem feita via dotnet DevOpsWebApi.dll --health-check.
```

Docker HEALTHCHECK devops-api-prod

OK

Criar compose.yaml com healthcheck sem curl · 2026-10-28 09:52:06

```
State.Health.Status = healthy; checagem feita via dotnet DevOpsWebApi.dll --health-check.
```

Containers do laboratorio apos limpeza

OK

Criar README, registrar evidencias e efetuar limpeza controlada · 2026-10-28 09:52:14

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe ps -a --filter name=devops-api
CONTAINER ID   IMAGE                COMMAND                  CREATED          STATUS                    PORTS
NAMES
e86d37b75236   devops-webapi:1.1.0   "dotnet DevOpsWebApi…"   14 seconds ago   Up 14 seconds (healthy)   127.0.0.1:8082-
>8080/tcp   devops-api-prod
7684163fcc79   devops-webapi:1.1.0   "dotnet DevOpsWebApi…"   14 seconds ago   Up 14 seconds (healthy)   127.0.0.1:8081-
>8080/tcp   devops-api-staging
```

Imagens preservadas 1.0.0/1.1.0/1.2.0

OK

Criar README, registrar evidencias e efetuar limpeza controlada · 2026-10-28 09:52:15

```
C:\Program Files\Docker\Docker\resources\bin\docker.exe image ls devops-webapi --format {{.Repository}}:{{.Tag}} {{.ID}}
{{.Size}}
devops-webapi:1.2.0 c259cbb5f566 342MB
devops-webapi:1.1.0 cf4728fe3520 342MB
devops-webapi:1.0.0 32e9d3541203 342MB
```

Requisitos do professor

OK

Criar README, registrar evidencias e efetuar limpeza controlada · 2026-10-28 09:52:15

```
Dockerfile, .dockerignore, compose.yaml, imagens versionadas, staging, production, healthchecks, falha intencional,
rollback e README foram gerados e verificados.
```

Relatorio elaborado a partir das saidas efetivamente registradas pelo script CEUB Docker Unidade 3.